

RESULTS

Results

01 · WILCOXON SIGNED-RANK

nonparametric paired comparison

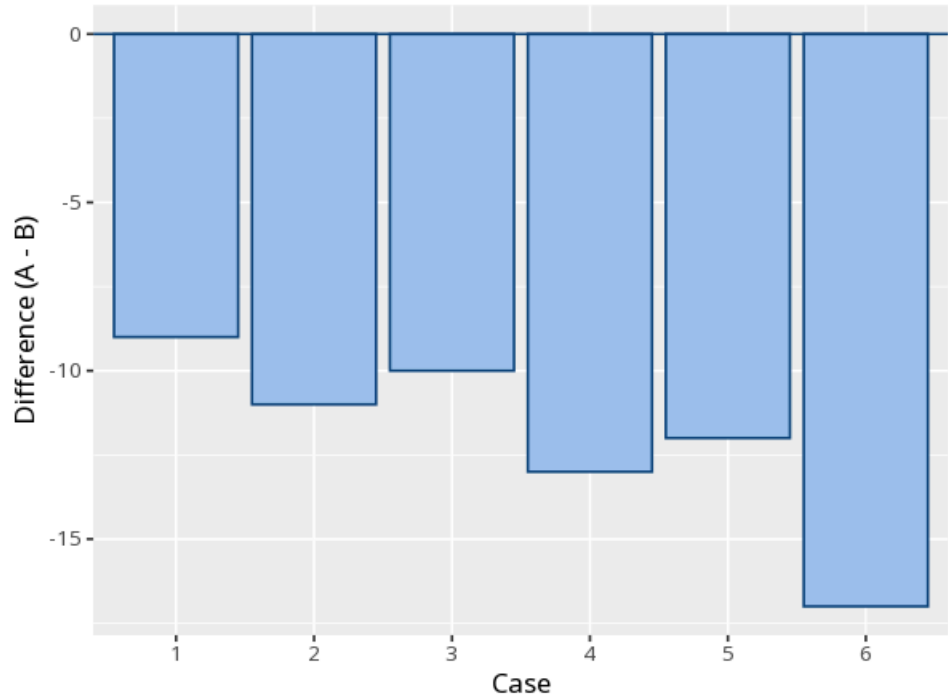
Table 1. Rank summary

Sign	N	Mean rank	Sum of ranks
Positive	0	—	—
Negative	6	3.50	21.00
Ties	0	—	—

Table 2. Signed-rank test

V / W	Z	p	r [95% CI]	Median diff [95% CI]
0.00	-2.20	.031	-1.00 [-1.00, -1.00]	-11.50 [-17.00, -9.00]

Figure. Change per case



Understanding the numbers

Hodges–Lehmann median diff. The estimated median of the paired differences (a location-shift estimate), with its confidence interval. Here, the Hodges–Lehmann median difference is -11.50 [$-17.00, -9.00$].

Mean rank. The average rank of the absolute differences within the positive- and negative-change groups. Here, the mean rank is — for positive changes and 3.50 for negative changes.

N. How many cases showed a positive change, a negative change, or no change (a tie). Here, $N = 0$ positive, 6 negative, and 0 tied.

p. The probability of a shift this large (or larger) if there truly were no systematic difference between conditions. Here, $p = .031$ - below the conventional .05 threshold, a significant shift.

r. The rank-biserial effect size for the shift between the two conditions. Here, $r = -1.00$ [$-1.00, -1.00$].

Sign. Whether each case's change was positive, negative, or tied (no change). Here, this run's cases split into 0 positive, 6 negative, and 0 tied changes.

Sum of ranks. The total of the ranks assigned to the positive and negative differences – the test statistic V is derived from this. Here, the sum of ranks is — for positive changes and 21.00 for negative changes.

V. The Wilcoxon signed-rank test statistic, the smaller of the summed positive/negative ranks. Here, $V = 0.00$.

Z. The standardized test statistic used to compute p . Here, $Z = -2.20$.

How to read this test

The nonparametric counterpart to the paired t -test. It ranks the size of each paired change; a p below alpha means a systematic shift between the two conditions, with r as the effect size. The shift maps to the median difference only when the within-pair differences are roughly symmetric; the sign of the median difference (or Hodges–Lehmann estimate) gives the direction. Your significance threshold (α) is 0.05. Method: Wilcoxon signed rank exact test.

APA template: A Wilcoxon signed-rank test gave $V=0.00$, $Z=-2.20$, $p = .031$, $r=-1.00$ [$-1.00, -1.00$].

Statistical basis: Wilcoxon signed-rank test (Wilcoxon, F. (1945). "Individual comparisons by ranking methods." *Biometrics Bulletin*, 1(6), 80-83.); Matched-pairs rank-biserial effect size (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." *Journal of Open Source Software*, 5(56), 2815.). Full references in the exported CITATIONS.txt.

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